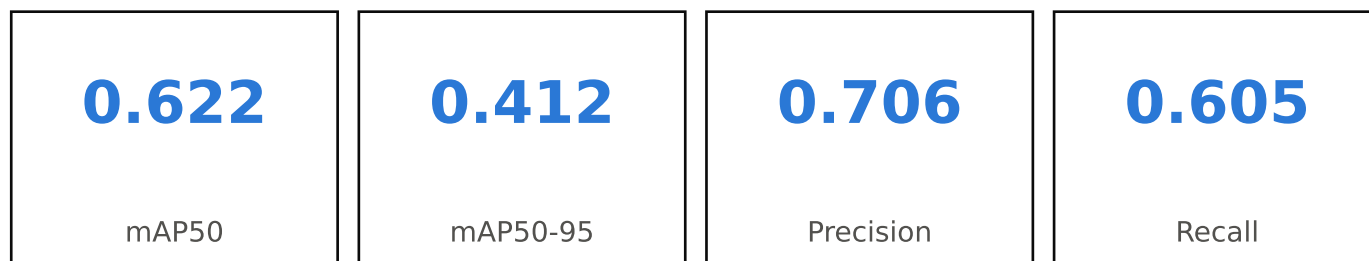
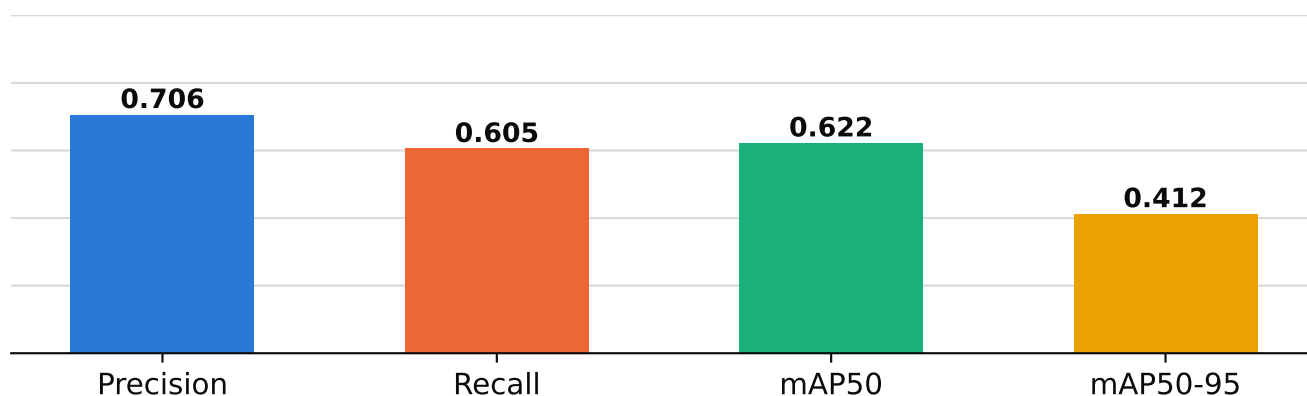


Document Layout Detection — Evaluation Report

RT-DETR-L fine-tuned on DocLayNet, evaluated on the held-out test split



Test set: 499 pages, 6,349 annotated regions, 11 classes (none in COCO)



Training configuration

Epochs: 15 Batch: 8 Image size: 640px

Optimizer: AdamW LR: 0.0001 Seed: 42

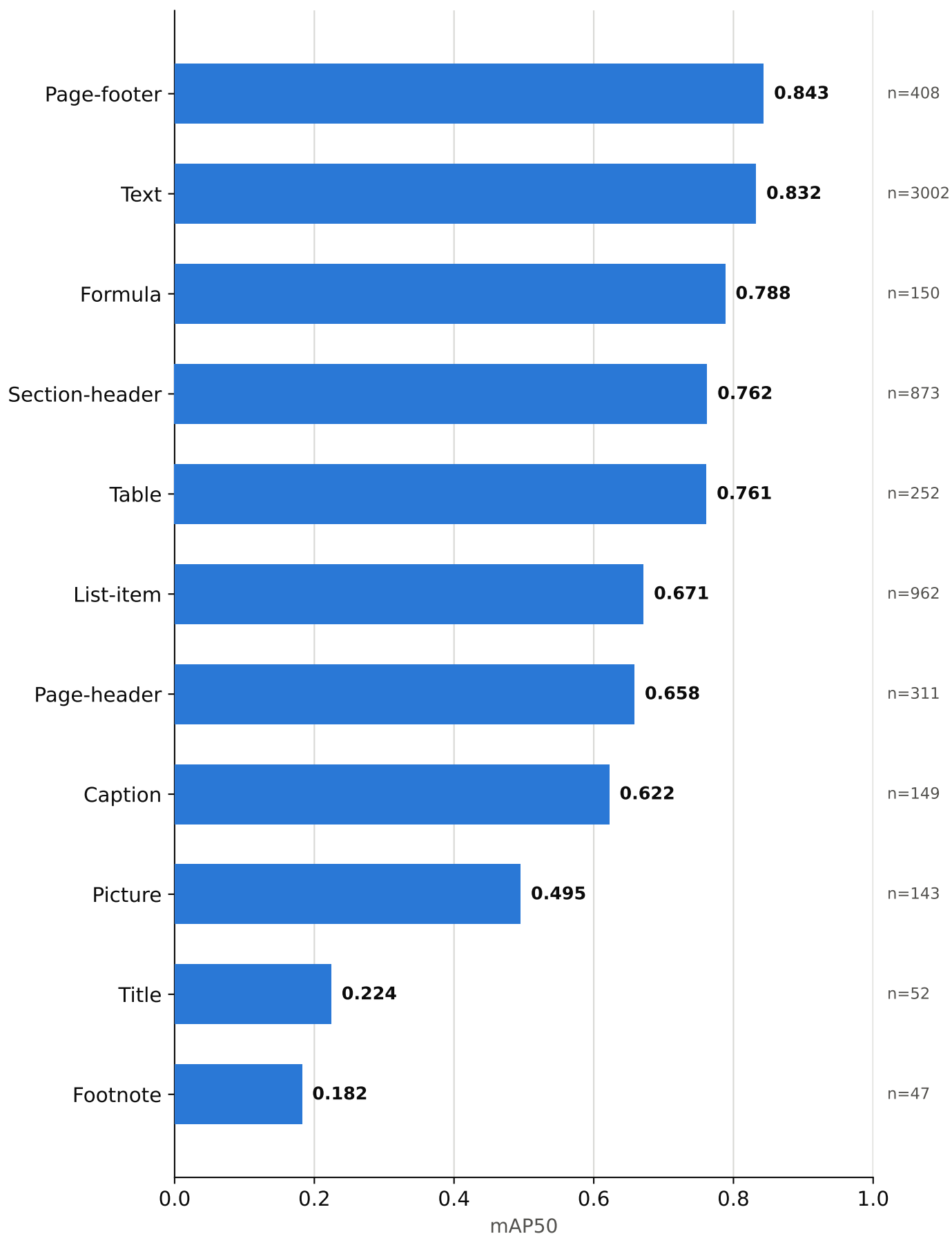
Hardware: Tesla T4 (14.6 GB) Wall clock: 2.27 h

Query-budget saturation: 0.0% of test pages (max 95 regions, budget 300)

Train/test source-PDF overlap: 0.0% (499 test pages checked)

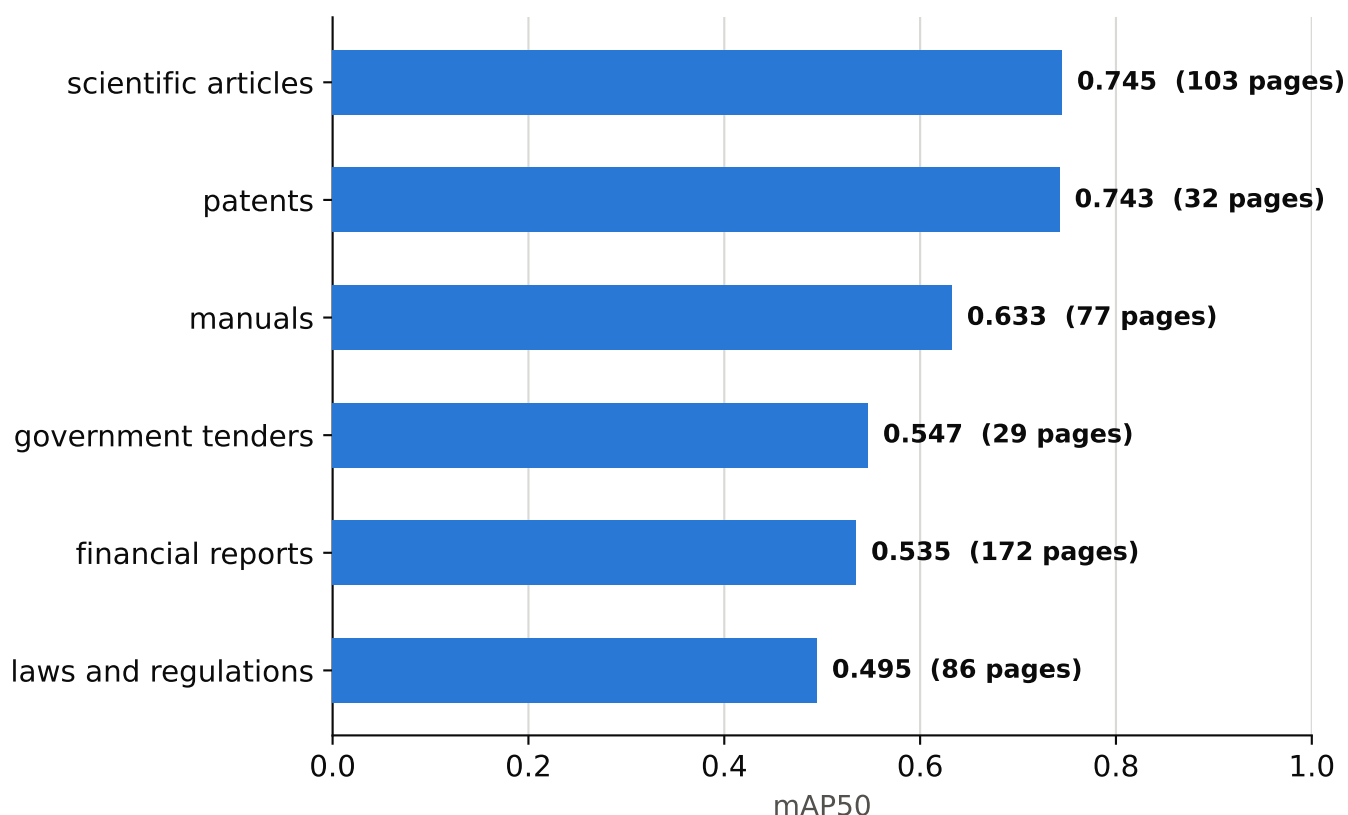
Per-class performance (mAP50)

Sorted by score. Instance count shown per class — rarity, not size alone, drives the weakest results.



Per-document-category performance (mAP50)

Tests generalization across document styles, not just aggregate accuracy.



Observed failure patterns

Footnote (47 instances): recall 0.048 — rare class, near-total miss rate.

Page-footer (408 instances): mAP50 0.843 — same thin shape as Footnote, but

9x more training instances and a consistent position. Rarity, not size,

is the dominant factor.

Picture: precision 0.416 despite recall 0.622 — composite regions (diagrams

with embedded text) get split into multiple overlapping boxes instead

of one region.

Dense repeated-entry layouts (directories, org charts) produce duplicate,

overlapping box proposals rather than one box per entry.